Performance Test

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	5 Marks

Project Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	**Data Rendered** **(Tables Loaded, Records Count and Size)**	• Database: HEALTH360_PATIENT_DB • Tables Loaded: 3 Base Tables (CLINICAL_RECORDS, DEMOGRAPHIC_RECORDS, LIFESTYLE_RECORDS) • Total Records Count: N = 5,000 Patient Cohort Records • Table Data Size: 2.4 MB (Compressed Snowflake Storage) • Status: 100% Data Successfully Rendered & Accessible
2.	**Data Preprocessing & Quality** **(Null checks, duplicate checks, trimming, standardization and data cleaning)**	• Null Check Audit: Initial nulls detected in BMI (124), BP (87), Glucose (95); resolved using Median/Mean Imputation. • Duplicate Check: 2 duplicate patient rows detected and removed using ROW_NUMBER() window partitioning. • String Standardization: Trimmed whitespaces & UPPER CASE conversion on PATIENT_ID. • Flag Categorization: Standardized boolean flags (0/1) for Diabetes, Heart Disease, and Stroke.
3.	New Fields Created (Any New Fields created)	• BMI_Category: Categorized into 'Underweight', 'Normal', 'Overweight', 'Obese' • Age_Group: Binned into 'Young Adult', 'Middle Age', 'Senior', 'Old' • Age_Group_Range: Binned into '18-29', '30-44', '45-59', '60-74', '75+' • Exercise_Category: Standardized into 'Level 0', 'Level 1', 'Level 2'. • Risk_Factor_Count: Derived sum of (Diabetes_Flag + Heart_Disease_Flag + Stroke_Flag)
4.	Filter Efficiency Test (Filters used + comparison of query performance with different filters)	• Global Filters Applied: BMI_Category, Age_Group, Smoking_Status, Exercise_Category • Unfiltered Query Execution Time: 120 ms (Full Table Scan across 5,000 records) • Filtered Query Execution Time (BMI_Category = 'Obese'): 45 ms (Snowflake Micro-partition Pruning applied) • Query Performance Gain: ~62.5% reduction in latency using indexed categorical filters
5.	Dashboard & Visualization Testing (No of Visualizations / Graphs)	• Total Visualizations / Graphs: 8 Interactive Tableau Story Views 1. Disease Prevalence Comparison (Bar Chart) 2. Patient Age Trend (Line Chart) 3. BMI VS Heart Disease (Bar Chart)

		4. Age Group Bubble Chart (Packed Bubble Chart) 5. Age Disease HeatMap (2D Intensity Matrix) 6. Clinical Scatter Plot (BP vs Glucose Scatter) 7. Exercise-level Pie Chart (3-slice Pie) 8. Smoking Status Analysis (Comparative Bar Chart) • Total Interactive Dashboards Built: 1 Main Storyboard ('Smart HealthCare And Risk Factor Analysis')
6.	Query History (Query with Highest Execution time)	• Query with Highest Execution Time: Multivariable Scatter Matrix Query (Clinical Scatter Plot join across clinical and demographic features) • Execution Time: 210 ms • Query SQL: SELECT Patient_ID, Blood_Pressure, Glucose, Heart_Disease_Status, BMI FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS WHERE Blood_Pressure IS NOT NULL AND Glucose IS NOT NULL; • Warehouse Resource Utilization: COMPUTE_WH (Size: X-Small, 0.002 credits used)